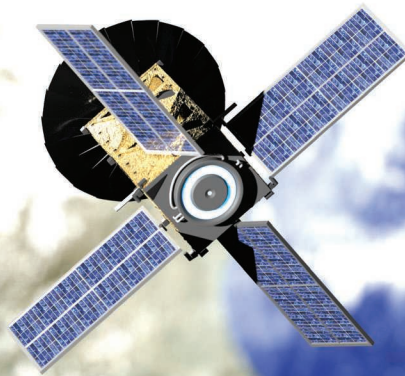


CubeSat Link Budget

Elements, calculations, and examples.

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Antenna designs suitable to fly on CubeSats have been of growing interest. The prolific CubeSat antenna literature, however, is contrasted by limited tools for an antenna engineer to develop or understand a CubeSat link budget, which sets the requirements for the spacecraft and ground antenna design. This article presents detailed explanations of the link budget basics, which include not only antenna design terms but also signal processing and system noise parameters. Relations among the data rate, signal-to-noise ratio (SNR), system noise temperature, and antenna gain are reviewed, explained, and applied in three examples of link budgets for CubeSats in low-Earth (LEO), geosynchronous (GEO), and lunar orbits.

INTRODUCTION

CubeSat, a modular type of standardized modern small satellite, has become increasingly popular since its introduction [1]. With advancements in digital signal processing, power electronics, and packaging technology, it is feasible to fit on CubeSats scientific instruments and communication devices that were traditionally carried on larger satellites. In addition, CubeSats can be launched to form constellations to distribute a large scientific mission into multiple smaller tasks. This is extremely attractive for space exploration because such a constellation of CubeSats not only reduces the mission cost, repair, and risk, but it also provides more precise and real-time science data. One may relate this to in situ probing of a biological tissue, where a smaller probe and more places to measure provide more accurate data. “The rise of the CubeSat,” as coined by *Science* [2], indeed describes the growth of CubeSat research and industry,

which is evident from Figure 1 and Table 1 [3]. The special issue on small satellites in the *IEEE Proceedings* [4] has a wealth of information on CubeSats. Readers are referred to it for details on CubeSat developments as well as terms such as *small satellite* (*SmallSat*) and nanosatellite (*NanoSat*).

Proportional to the rise of CubeSats, antennas—critical components of CubeSats—have also received much due interest. This is evident from the number of papers and special issues on the topic [5], [6]. A CubeSat antenna may provide some or all of the following functions: telemetry tracking and command (TT&C), high-speed downlink for payload data, receiving positioning data, and intersatellite cross links [7]. For more details on the terminology, readers are referred to a space mission design text book [8] and lecture notes published by the Massachusetts Institute of Technology [9], which includes excellent graphical explanations of satellite communication links and TT&C.

Depending on its mission, a CubeSat may require some or all of these antenna capabilities. Some of them could be realized by one antenna, such as using the same antenna for TT&C and downlink or determining spacecraft attitude without utilizing a GPS antenna, as presented in [10]. However, most often, different antennas are required to keep the CubeSat assembly in a modular fashion. Therefore, interaction and crosstalk between different antennas could become an area of study. On the other hand, antenna engineers strive to develop antennas that can pack more functionality into one multiband and diversified antenna unit.

While there is a large body of work on CubeSat antennas, the link budget, a comprehensive consideration of antenna design and signal processing, seems to have limited discussions and clarifications in the antennas and propagation community.

It is often discussed in the communication, signal processing, and system design literature [11], [12], [13], [14]. Reference [15] discusses the link budget in great detail, but it may be intimidating for some antenna engineers.

A basic capability to understand, read, and modify a CubeSat link budget is not only an important skill set for antenna engineers but also a guideline to determine the requirements (e.g., the frequency, gain pattern, and bandwidth) for the CubeSat antenna design. Accordingly, the objective of this article is to provide the fundamentals of developing a CubeSat link budget in language toned for antenna engineers.

OVERVIEW OF A CUBESAT LINK BUDGET

A CubeSat ↔ ground link, as illustrated in Figure 2, is generally studied through three main composites: the transmitter (TX), wireless path, and receiver (RX). This applies to the TT&C

TABLE 1. NANOSAT FACTS AS OF 1 JANUARY 2021 [3].	
NanoSats launched	1,474
CubeSats launched	1,357
Interplanetary CubeSats	2
NanoSats destroyed on launch	93
Most NanoSats on a rocket	103
Countries with NanoSats	70
Forecast for the next six years	More than 2,500 NanoSats to launch

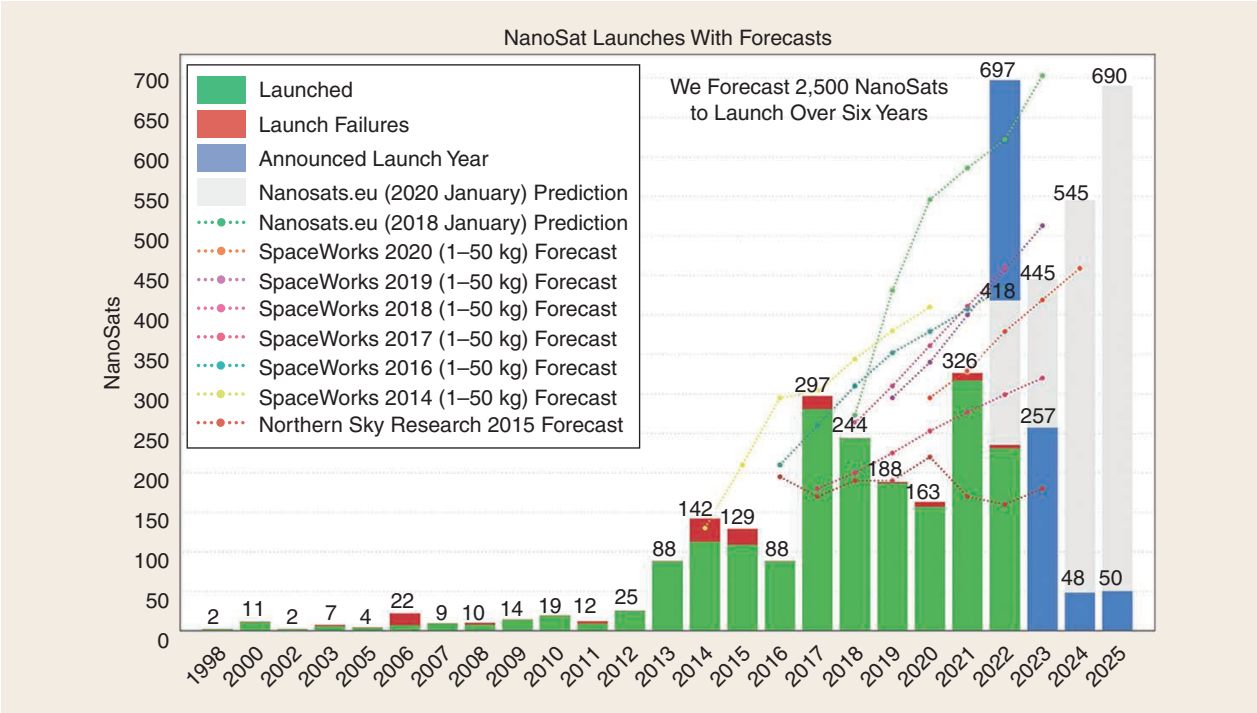


FIGURE 1. NanoSat launch data. (Source: [3]; used with permission.)

ground-to-satellite uplink and the CubeSat-to-ground downlink. For example, in an uplink, the ground station is the TX and the RX is on the spacecraft. In Figure 2, the same CubeSat antennas are used for both the telemetry and downlink. A block diagram for the TX and RX is illustrated in Figure 3. From the TX to RX, the transmitted signal undergoes power loss, listed as follows:

- *Loss in the TX before reaching the TX antenna:* This includes loss in transmission lines (i.e., lines A, B, and C in Figure 3); insertion loss of the filter; and other inline devices before the antenna, such as a directional coupler.
- *Path loss and atmospheric loss:* This includes the decay of the spherical wave due to the distance between the TX and RX antennas, loss due the electromagnetic wave traveling through atmospheric gases (nitrogen, oxygen, carbon

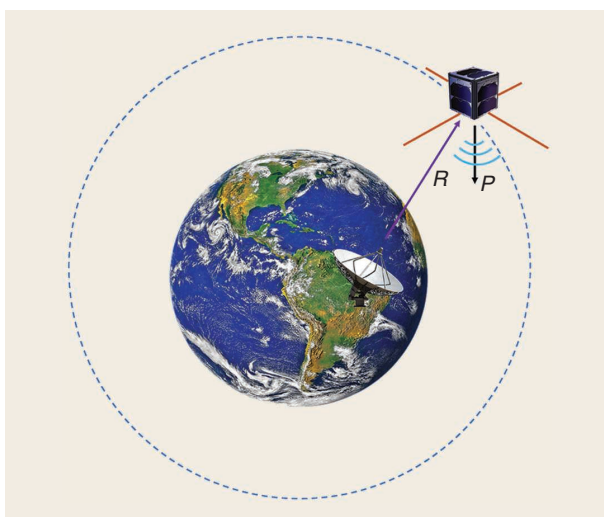


FIGURE 2. An illustration of a CubeSat ↔ ground link.

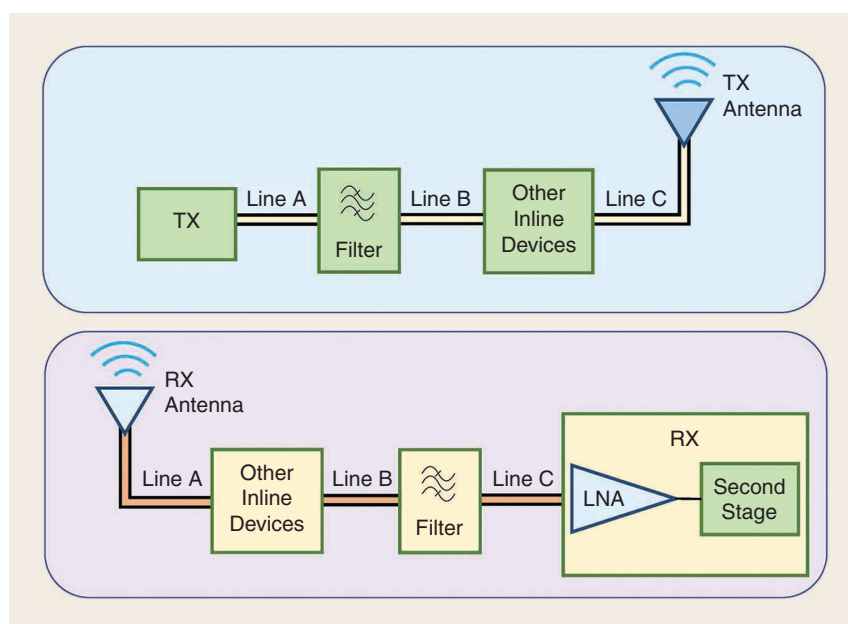


FIGURE 3. A block diagram of TX and RX front ends.

dioxide, hydrogen, etc.), and the attenuation when the electromagnetic wave travels through the ionosphere. These losses often are summed together as the link loss.

- *Loss in the RX after the signal is intercepted by the RX antenna:* This includes loss in the transmission lines and/or waveguides; insertion loss of the bandpass filter; and other inline devices before the low-noise amplifier (LNA), such as a diplexer or hybrid.
- *Polarization mismatch loss:* This occurs when the polarization of the RX and TX antennas are out of alignment. One solution is to use antennas with the same circular polarization for the spacecraft and ground station.
- *Pointing loss:* From Figure 2, it is obvious that the direction of the maximum gain spacecraft antenna **P** is not inline with the pointing of the ground antenna **R**. Such misalignment is estimated and included in the link budget as the pointing loss. It should be noted that the ground station is designed to rotate and track the satellite. Hence, it points to the satellite, and pointing loss from the ground antenna is normally small and easily estimated. The pointing loss of the spacecraft antenna is determined from the location of the satellite on an orbit, its orientation, and its gain pattern. Most spacecraft system or instrument engineers have calculators to determine the angle between **P** and **R** for a specific location of the satellite. Using this angle and the spacecraft antenna's gain pattern, the pointing loss can be estimated. For example, for a half-wavelength dipole antenna without a ground plane, the pointing loss at 45° from its boresight is 3 dB.

The RX antenna, in addition to receiving the signal from the TX antenna (i.e., the TX signal minus all of the losses discussed), also receives noise from its environment—or, in other words, the RX antenna is heated by the sources that it is pointing at (e.g., the sky). In addition, as long as the RX antenna is not 100% efficient, it generates heat due to its loss resistance. Similarly, there is heat generated in transmission lines (lines a, b, and c in Figure 3) and in the RX. Together, these heat sources account for the noise in the RX.

The ratio between the power that reaches the RX and the noise in the RX is the signal-to-noise ratio (SNR), and it has to be larger than a threshold value in order for the RX to recover the signal. For a CubeSat ↔ ground link, there is a required SNR that will be discussed in more detail in the next section, and one calculates the link margin from (1), where S/N is the SNR at the RX, and the values are in decibels:

$$\text{Link Margin} = \frac{S}{N} - \left(\frac{S}{N} \right)_{\text{required}} \text{ (dB)}. \quad (1)$$

When the link margin is ≥ 0 dB, the term *link closes* is used, meaning that the RX could separate and process

the signal from the noise floor. In general, one prefers the link margin to be higher than 3 dB.

As a section summary, a CubeSat link budget is a calculation of the link margin, i.e., how much SNR is needed at the RX end for a desired data rate. The entries that go into the calculation are desired to be as inclusive and accurate as possible. Some of the budget parameters are statistical, and, therefore, the budget is an estimation technique for evaluating the communication link performance. In that sense, a generous link margin is desired to ensure an effective communication when factors of a random nature may occur.

BASIC ELEMENTS OF A LINK BUDGET

A CubeSat link budget is a comprehensive matrix that includes transmitted power; detailed signal attenuation; noise sources; and the required SNR, which is determined from the signal processing method employed in the link. The major entry to the link loss (excluding atmospheric loss) and the effect of the TX and RX antenna parameters are computed from the Friis transmission equation. The SNR is defined and computed from the signal processing method. The noise power is computed from the system noise temperature. These are basic elements of a link budget, and a detailed explanation is presented in this section.

Although tabulation and computer codes are reasonable methods for presenting or computing a link budget, CubeSat teams often make an extensive Excel workbook that includes the entry of communication details. The name *link budget* is often used for such a workbook in the CubeSat community. The link budget developed by Jan King is a very comprehensive workbook and is a great resource for those who are interested in starting a CubeSat program [16]. Also, a CubeSat link budget can be made a lot simpler by entering estimated values rather than detailed calculations or by eliminating some factors, as shown in the “Common Simplifications Employed in Link Budget Developments” section.

FRIIS TRANSMISSION EQUATION AND LOSSES

The Friis transmission equation is a main entry in a CubeSat link budget. An extended version that includes a mismatch between the antenna and transmission line as well as a polarization mismatch can be written as follows [17]:

$$\frac{P_r}{P_t} = \left(\frac{\lambda}{4\pi R_L} \right)^2 (1 - |\Gamma_r|^2)(1 - |\Gamma_t|^2) G_r G_t |\rho_r \cdot \rho_t|^2, \quad (2)$$

where the symbols are as follows:

- P_r is the power at the RX antenna to be passed on to line a (Figure 3) (W).
- P_t is the power at the input port of the TX antenna (W).
- λ is the wavelength (m).
- R_L is the distance between the CubeSat and the ground antenna, also called the *path length* (m).
- Γ_r and Γ_t are the reflection coefficients between the antenna and the feed line at the RX and TX, respectively.
- G_r and G_t are the gain pattern of the RX and TX antennas, respectively.
- ρ_r and ρ_t are the polarization of the RX and TX antennas, respectively. $|\rho_r \cdot \rho_t|^2$ calculates the polarization mismatch

loss. When the two antennas match in polarization, then the polarization efficiency is 100%, and the loss is 0.

To construct a link budget, entries and calculations on the TX side can be as in Table 2. The line attenuation constant is the attenuation of cables and waveguides. The attenuation in the transmission line is $e^{2\alpha l}$ because it is the attenuation of power (not voltage or current, which is $e^{\alpha l}$).

Accordingly, (2) is converted to dB form to compute P_r . Note that the link loss is a summation of path loss (sometimes called the *space loss*) $L_s = 20 \log_{10}[\lambda/(4\pi R_L)]$ and atmospheric losses.

The power that reaches the RX, P_{RX} , which is the signal power to compute the SNR, can now be calculated by P_r minus the total loss on the RX side. Tabular entries for the RX can be created in a similar fashion as was done for the TX.

THE SNR AND LINK MARGIN

It is noteworthy to have a short discussion on the ratio E_b/N_0 when constructing a link budget. In the ratio, E_b is the signal energy per bit (J/bit), and N_0 is the spectrum noise density (W/Hz). One could easily show, by verifying units, that E_b/N_0 is, indeed, the SNR per bit. For a modulated signal, we are interested in the carrier-to-noise ratio (CNR), often written as C/N . The CNR is the SNR of the received signal, after the RX filter but before detection. One could then readily show the following:

$$\frac{C}{N} = \frac{P_{\text{signal}}}{N} = \frac{E_b \cdot R}{N_0 \cdot B} = \left(\frac{E_b}{N_0} \right) \cdot \frac{R}{B} \quad (3)$$

where R is the channel data rate (bit/s), and B is the bandwidth (Hz).

The minimum E_b/N_0 threshold required for a modulated channel is related to the modulation method and bit error rate (BER). Figure 4 shows the relation of BER versus E_b/N_0 for a few different modulation methods. Examples of modulation methods include binary/quaternary phase-shift keying (BPSK/QPSK); Gaussian minimum-shift keying (GMSK); and a coded modulation, such as QPSK with convolutional (conv.) code. This

TABLE 2. LINK BUDGET CONSTRUCTION: ENTRIES FOR TX.

TX power	P_{TX} (dB)
Length of cables and waveguides	$L = \Sigma$ length of lines A, B, and C
Line attenuation constant	α
Total connector loss	L_c
Filter insertion loss	IL
Directional coupler loss	L_{coupler}
Line loss	$L_l = 10 \log_{10}(e^{2\alpha l})$ (dB)
Total loss	$L_T = L_l + L_c + IL + L_{\text{coupler}}$ (dB)
P_t	$P_{TX} - L_T$ (dB)

figure was generated by the author using the BER tool in MATLAB. One could also get a similar graph by following a digital communication text [18]. It is clear from the figure that, for less error in a channel, one needs a higher E_b/N_0 . For a given modulation and BER, the corresponding E_b/N_0 is the threshold required in the communication link.

LINK MARGIN: E_b/N_0 METHOD

To compute the RX C/N , one simply needs to have P_{RX} divided by the noise power in the RX. The noise power is computed from the noise power density that is caused by the RX system temperature, explained in the “Noise and System Noise Temperature” section. Labeling this noise power density as N_{sys} , we could rewrite (3) into

$$\frac{E_b}{N_0} = \frac{C}{N} \cdot \frac{B}{R} = \frac{P_{RX}}{N_{sys} \cdot B} \cdot \frac{B}{R} = \frac{P_{RX}}{N_{sys} \cdot R}. \quad (4)$$

The link margin is then obtained from the difference between the computed RX E_b/N_0 (converted to dB) and the threshold, which is decided by the BER and modulation method. For example, in Figure 4, if BPSK is chosen, and the maximum acceptable BER is 10^{-5} , then the E_b/N_0 threshold is 9.6 dB. Accordingly, the formula for the link margin is expressed as follows:

$$\text{Link Margin} = \frac{E_b}{N_0} - \left(\frac{E_b}{N_0} \right)_{\text{threshold}} \quad (\text{dB}). \quad (5)$$

It is clear from (4) that, to maintain an adequate link margin (e.g., 3 dB) for a higher data rate at a given TX power, one needs

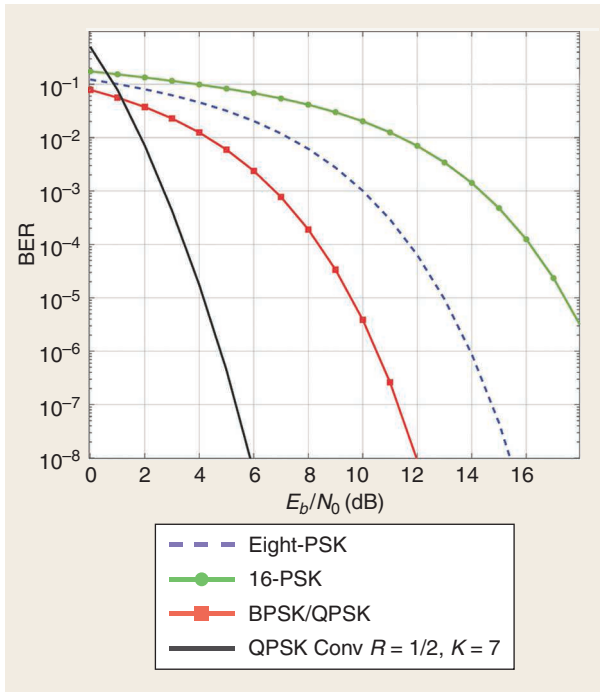


FIGURE 4. The E_b/N_0 versus BER for different modulations.

a higher P_{RX} , which then means a need for high-gain antennas [from (2)].

CODED CHANNEL: E_b/N_0 THRESHOLD AND CODING GAIN

It is known in communication theory that the channel capacity (bit/s) for a coded channel is governed by Shannon's theorem:

$$R_c = B \cdot \log_2 \left(1 + \frac{S}{N} \right) \quad (6)$$

where R_c is used to label the channel capacity in this article to be consistent with the previous notation for data rate. The ratio S/N is the SNR of the coded channel.

Substituting S/N as computed from P_{RX} and N_{sys} (as in the “Link Margin: E_b/N_0 Method” section), we reach

$$R_c = B \cdot \log_2 \left(1 + \frac{P_{RX}}{N_{sys} \cdot B} \right) \quad (7)$$

which says that, in addition to high-gain antennas, a wider bandwidth promises a higher data rate since the $\log_2$ function varies slower than $y = x$ line.

One could interpret Shannon's theorem in application as meaning that the E_b/N_0 threshold of a channel is reduced by coding. In Figure 4, the curve of QPSK with conv. code is compared to uncoded QPSK. As seen, the E_b/N_0 threshold for a BER of 10^{-6} is 10.5 dB and 4.8 dB for uncoded and coded QPSK, respectively. Some link budget developers call the difference the *coding gain*.

LINK MARGIN: S/N METHOD

It is apparent that the E_b/N_0 method does not apply to analog channels. For analog systems, the SNR is calculated from $P_{RX}/(N_{sys} \cdot B)$, and the link margin is then computed from (1), where the required SNR is determined by the communication system. For example, for a simple single-frequency, two-antenna system (i.e., a TX and an RX antenna), an RX may only require P_{RX} to be 3 dB higher than the noise floor. For a modulated analog signal, noise power due to the modulation and demodulation needs to be added to the noise floor, and those numbers are well defined and calculated in communication text books.

It is a common practice for link budget developers to term the procedure described in (8) as the *S/N method*, where the SNR at the RX is computed, and the threshold is taken to be the same as the E_b/N_0 threshold:

$$\text{Link Margin} = \frac{P_{RX}}{N_{sys} \cdot B} - \left(\frac{E_b}{N_0} \right)_{\text{threshold}} \quad (\text{dB}). \quad (8)$$

It is straightforward to verify that the link margin computed from the S/N method is smaller than that from the E_b/N_0 method for channels where the ratio B/R is bigger than 1, which is the case for most uplinks. Therefore, many link budget analysts start with the E_b/N_0 method to develop their budget since the bandwidth information is not needed and then add on the S/N link margin method as an additional measure for the link closure.

NOISE AND SYSTEM NOISE TEMPERATURE

Electrical noise is associated with temperature, and the noise power density is calculated from $k_B \cdot T_N$, where $k_B = 1.380649 \times 10^{-23}$ (J/K) is the Boltzmann constant, and T_N (K) is the noise temperature of a noise source. The subscript N is added to differentiate the noise temperature from the physical temperature T_P (K).

The system noise power density in the RX is contributed from the following noise temperatures:

- **Antenna noise temperature T_A :** This accounts for when the antenna is heated up by receiving radiation from the surrounding sources, such as the sky, ground, bright stars, etc.
- **Antenna loss temperature:** Heat due to the loss resistance of the antenna. While the T_A is a temperature heated by external factors, the loss temperature of an antenna T_{AL} is an internal heat generated on the antenna.
- **Noise temperature of transmission lines, labeled as T_{LL} :** This is due to the unit length resistance and conductance of the lines and is clearly related to the attenuation of the line.
- **RX noise temperature, labeled as T_R :** This is related to the heat generated inside the RX and is calculated from the noise temperature and gain of stages in the RX (the LNA, second amplifier stage, etc.) [19].

Together, these noise temperatures account for the total system noise temperature T_{sys} of the RX system, which includes the RX and front end. Accordingly, the RX system noise density is computed from $N_{sys} = k_B \cdot T_{sys}$.

T_{sys} CALCULATIONS

From discussions in the previous section, it is seen that we need T_A , T_{AL} , T_{LL} , and T_R to compute T_{sys} . Derivations of T_{sys} have been well discussed and explained using system and functional analysis methods [20], [21]. This article intends to provide a derivation in a conceptual sense that may be easier to grasp for antenna engineers.

ANTENNA TEMPERATURE

The antenna noise temperature is calculated as follows [22]:

$$T_A = \frac{1}{\Omega_A} \int_0^\pi \int_0^{2\pi} T_s(\theta, \phi) P_n(\theta, \phi) \sin(\theta) d\theta d\phi \quad (9)$$

where

- T_A is the antenna noise temperature (K)
- $T_s(\theta, \phi)$ is the brightness temperature of sources as a function of angle (K)
- $P_n(\theta, \phi)$ is the normalized antenna power pattern (dimensionless)
- Ω_A is the antenna beam solid angle (sr).

As seen, the antenna temperature is related to the directional properties of the antenna as well as the source temperature. For example, if a ground antenna is highly directive and the minor lobe level very low, then the antenna temperature is only affected by the sky temperature. Otherwise, both the sky temperature and ground temperature, commonly taken as 290 K [22], need to be entered in (9).

Temperature of the cold sky is 2.7 K, but there are bright sources that raise the sky temperature [22]. One of them is the galactic noise temperature, as it can be substantial for frequencies below 1 GHz. The galactic noise temperature (T_G) is calculated from (10) [23]. In (10), f is the frequency of the antenna, and t_c is the temperature of the contour drawn by Kraus [24]. The coldest and hottest galactic noise temperatures are found by letting t_c equal 0 and 50, respectively. It should be noted that the value 50 was not marked on the radio sky map [24]. It is extrapolated by studying the existing numbers on the map.

$$T_G = (6 \times t_c + 80) \left(\frac{f}{0.25 \times 10^9} \right)^{-2.75} + 2.7. \quad (10)$$

In practice, the antenna temperature can be either estimated with simplifications (see the “Common Simplifications Employed in Link Budget Developments” section) or measured (for the ground antenna). Measuring the noise temperature of the ground antenna can be done as follows [16]. One could connect a spectrum analyzer or the RX’s signal strength meter directly to the end of the antenna. The noise power measured on the spectrum analyzer or the meter is the terrestrial noise plus the noise floor of the analyzer (or meter). The noise floor can be determined by replacing the antenna with a 50-Ω load and then reading the power level. This way, subtracting the noise floor value from the measured total noise power level gives the antenna noise temperature, which includes T_A and the internal loss temperature. It is possible that the terrestrial noise power level is not considerably higher than the meter’s noise floor level, which means no detectable antenna temperature. In that case, noise power can be set as being equal to or less than the meter’s noise floor level. This number can then be used to compute the antenna temperature by dividing it by k_B and the bandwidth of the meter.

ANTENNA AND TRANSMISSION LINE LOSS TEMPERATURE

Figure 5 shows an illustration of an RX antenna with a matching resistor R as its load. If the efficiency of the antenna is e_A , then the power delivered to the load is $e_A P_{in}$. The resistor takes this delivered power and dissipates it as heat. The physical temperature of the antenna, as it is connected to the resistor, is the same as the heat on the resistor and, therefore, can be calculated from $e_A P_{in} / (k_B B)$. The power loss on the antenna is $P_{in} - P_{out} = (1/e_A - 1)P_{out}$, which corresponds to

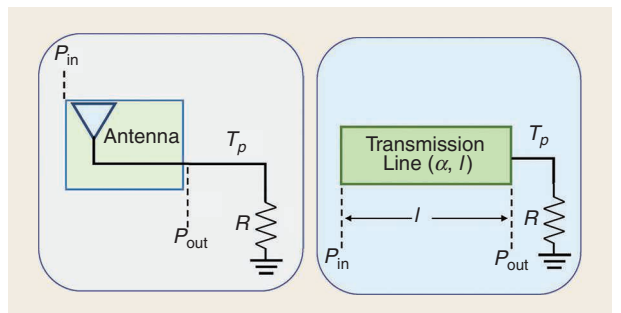


FIGURE 5. The noise temperature and physical temperature.

the loss temperature. Accordingly, the loss temperature of the antenna is

$$T_{AL} = \left(\frac{1}{e_A} - 1 \right) T_{PA} \quad (11)$$

where T_{PA} is the physical temperature of the antenna.

Using the same argument, the loss temperature of a transmission line can be computed from its physical temperature T_{PL} and attenuation from (12) by noting that $P_{out} = e^{-2\alpha l} P_{in}$ corresponds to the physical temperature and that the power loss ($P_{in} - P_{out}$) leads to the loss temperature:

$$T_{LL} = (1/e_L - 1) T_{PL} \quad (12)$$

where $e_L = e^{-2\alpha l}$ is the line efficiency, and l is the length of the line.

SYSTEM NOISE TEMPERATURE

To compute the system noise temperature, it is convenient to combine all transmission lines (Figure 3), filters, and other inline devices as one with a combined efficiency of e_L , as illustrated in Figure 6. Note that this e_L includes the transmission line efficiency e_l in (12). It is also important to specify the reference location where the system temperature is calculated. The system noise temperature at the antenna (i.e., rf_1 in Figure 6) is not the same as at the RX (i.e., rf_3). This article labels these two temperature as T_{sys}^A and T_{sys}^R , respectively.

The calculation of T_{sys}^R follows the same method in finding the noise figure of a cascaded system [19]. The antenna intercepts the terrestrial noise power, which is $k_B T_A B$, and then passes this power to its output after multiplying by the antenna efficiency. The antenna generates loss power $k_B T_{AL} B$ due to its loss temperature T_{AL} , and it needs to be multiplied by e_A as well. Therefore, the total noise from the antenna to the transmission line is $k_B e_A (T_A + T_{AL}) B$. This power plus the combined line loss temperature of $k_B (1/e_L - 1) B$ [one could reach this form with the same manner in getting (12)] reach the RX after being multiplied by e_L . Finally, the total system noise power at the RX is the summation of the noise power that reached the RX and the noise power generated by the RX $k_B T_R B$, where T_R is the RX noise temperature. Dividing the system noise power at the RX by the bandwidth and Boltzmann constant yields

$$\begin{aligned} T_{sys}^R &= e_A \left[T_A + \left(\frac{1}{e_A} - 1 \right) T_{PA} \right] e_L + e_L \left(\frac{1}{e_L} - 1 \right) T_{PL} + T_R \\ &= e_A e_L T_A + e_L (1 - e_A) T_{PA} + (1 - e_L) T_{PL} + T_R. \end{aligned} \quad (13)$$

The system noise temperature at the RX goes into (4) as $N_{sys} = k_B T_{sys}^R$ to compute E_b/N_0 .

Although T_{sys}^A is not needed in the link budget calculation, it is explained in this article as a reference. From the RX end, to output noise power $k_B T_R B$, the input power of the line has to be $k_B T_R B / e_L$. This value plus the internal loss power of the line is now the output noise power of the antenna. Using the same argument, the noise power at the input port (i.e., rf_1) is the terrestrial radiation received by the antenna, loss power of the antenna, plus $1/e_A$ times the power at its output. Accordingly, we reach

$$\begin{aligned} T_{sys}^A &= T_A + T_{AL} + \frac{1}{e_A} \left(T_{PL} + \frac{T_R}{e_L} \right) \\ &= T_A + \left(\frac{1}{e_A} - 1 \right) T_{PA} + \frac{1}{e_A} \left(\frac{1}{e_L} - 1 \right) T_{PL} + \frac{T_R}{e_A e_L}. \end{aligned} \quad (14)$$

Note that (14) is consistent with [20] and [21], and one should be careful when reading different texts, as notations could be different, and typos in noise temperature notes are not uncommon.

To complete all elements that contribute to T_{sys}^R , the noise temperature of the RX is copied from [19] as follows:

$$T_R = T_{LNA} + \frac{T_{\text{second stage}}}{G_{LNA}}. \quad (15)$$

As seen, noise temperatures of the LNA, second stage, and the gain (unitless value) of the LNA need to be entered into (15). The noise temperature of the LNA and second stage can be calculated from their respective noise figures [19].

COMMON SIMPLIFICATIONS EMPLOYED IN LINK BUDGET DEVELOPMENTS

It is fairly common for CubeSat teams to use simplified methods for estimating T_{sys} . For example, the physical temperature of the ground RX antenna and lines are often taken as 290 K, which is the accepted Earth temperature [22]. It is also acceptable to use the same temperature for the spacecraft RX, especially when the CubeSat is in LEO.

SIMPLIFIED ANTENNA NOISE TEMPERATURE

The source temperature T_s in (9) can be simplified as having no angular variation. This way, T_A can be approximated by using an average for the integral. If the CubeSat antenna is directional with low minor lobes and is facing Earth, then its noise temperature is 290 K. If the antenna has an omnidirectional pattern, then (9) is easily seen as being an average of the Earth and

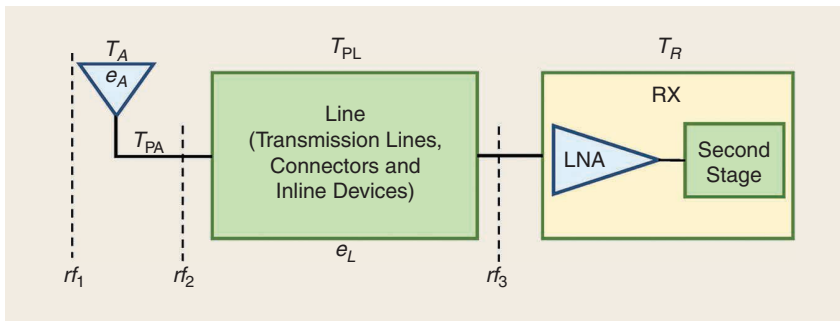


FIGURE 6. An illustration of system noise temperature.

sky temperature of 2.7 K. Therefore, some teams put 150 K, a number close to half of the Earth temperature for T_A [11], [12], [13], [14]. It should be noted that this 2.7 K does not include the galactic noise temperature, which is calculated from (10).

As seen, the galactic noise temperature is lower than 3 K for frequencies higher than 1 GHz, but it can be substantial for lower frequencies. This means that, when very-high-frequency (VHF) antennas are utilized in the communication link, one needs to be careful and take into account when and how long the antenna points to the hot region of the Milky Way. An example for galactic noise is shown in the “Link Budget of a GEO CubeSat” section.

For the ground antenna, T_A can be either measured as explained in the “ T_{sys} Calculations” section or estimated by taking an average of the Earth and sky temperature with galactic noise included for frequencies lower than 1 GHz. It should be noted that, although the ground antenna looks up at the sky, the back and side lobes pick up the Earth temperature; hence, taking half of 290 K is a safe estimation. Some teams may enter the Earth temperature as T_A for the ground antenna to leave room for any possible noise or loss that is not considered in the initial analysis stage.

SIMPLIFIED PHYSICAL TEMPERATURE AND RX TEMPERATURE

The physical temperature for the ground antenna, lines, inline devices, and RX is often set as the same as the Earth temperature. The same is often done for an LEO spacecraft’s RX system as a safe estimation. Some may enter a smaller number, such as 280 K, because of the temperature drop in space, especially at higher altitude orbits.

The RX temperature is computed from (15). Since the LNA has a moderately high gain, the contribution from the second stage is relatively low. It is acceptable to enter the LNA temperature with some margin as the RX noise temperature. For example, if the LNA gain is 15 dB, and if the second-stage temperature is 20 times the LNA temperature, which is relatively extreme, then the RX temperature is $1.63T_{\text{LNA}}$.

The noise temperature of the LNA and the second stage can be found from

$$T = (F - 1)T_P, \quad (16)$$

where F is the noise figure, and T_P is the physical temperature of the RX [19].

Some teams may calculate the spacecraft RX temperature from the RX sensitivity, which is included in the vendor data. This could be a case when the noise figures of the RX stages are unknown. RX sensitivity S_i is computed from

$$S_i = k_B T B \frac{C}{N} = k_B R \frac{E_b}{N_0} \quad (17)$$

where T includes the antenna and RX temperature, and the rest of the symbols are the same as in previous sections. Estimation

of the RX temperature using this method yields a higher number since the antenna temperature is included, or one could subtract the antenna temperature from T . An example in the “CubeSat Link Budget Examples” section uses this method to estimate the RX temperature and has a high but safe T_{sys} value for the link budget.

SIMPLIFIED SYSTEM TEMPERATURE

Some link budget developers may treat the efficiency of the antenna as 1 to simplify (13). When the antenna noise temperature and physical temperature of the RX (including the front end) are taken as the same (e.g., the Earth temperature), T_{sys} is then simplified to be the physical temperature plus the RX noise temperature.

APPLICATION NOTES

With the link budget elements and calculation methods explained in the preceding sections, it is worthwhile to point out several notes that are of practical interest.

RX FIGURE OF MERIT G/T

This is an important measure and is the ratio of the effective RX antenna gain (i.e., the gain minus the transmission line loss) over the RX system noise temperature. It is clear that, for the same antenna gain, a higher RX figure of merit means less system noise at the RX. In applications, when the RX figure of merit is known, one can compute the system noise temperature from the gain and G/T . There is also an antenna figure of merit (G_a/T_a). Most ground station vendors provide this number, and some link budget developers use this figure and the ground station antenna’s gain to estimate the antenna temperature and then enter it in (13) to calculate the system noise temperature.

DEMODULATOR IMPLEMENTATION LOSS

When entering the required E_b/N_0 , one should also include the demodulator implementation loss. Because a demodulator may not be perfect, the required E_b/N_0 value may be higher than the theoretical value (e.g., the one found from Figure 4). This difference is called the *demodulator implementation loss* and should be added to the theoretical E_b/N_0 threshold. A more precise implementation loss of an RX can be measured using a BER tester and a noise generator. In practice, however, entering 1–1.5 dB for combined hard- and software implementation loss is a reasonable estimation, considering the advances in digital communication technology.

OMNIDIRECTIONAL LOWER FREQUENCY UPLINK SPACECRAFT ANTENNA

A spacecraft uplink antenna is to receive a TT&C signal from the ground, preferably at any location—or at least most locations—on the orbit and at any spacecraft orientation. This includes when the satellite is tumbling so that the ground could command it to adjust itself. This means the pointing loss

TABLE 3. LEO LINK BUDGET.

Constants	Symbol	Value	Units
Speed of light	c	3E+8	m
Boltzmann constant	k_B	1.38E-23	J/K

Design Element	Symbol	Units	Uplink	Downlink
Frequency	f	GHz	0.45	0.465
TX power	P_{TX}	Watts	5	1
TX power	P_{TX}	dBW	6.99	0

TX	Symbol	Units	Uplink	Downlink
Antenna gain	G_{TX}	dB	37	-3.5
Total TX loss	L_{TX}	dB	0.5	0.5
Pointing loss	$L_{\theta TX}$	dB	0.44	0
Power transmitted	P_t	dBW	43.05	-4

RX	Symbol	Units	Uplink	Downlink
Antenna gain	G_{RX}	dB	-3.5	37
Total RX loss	L_{RX}	dB	0.5	0.5
Pointing loss	$L_{\theta RX}$	dB	0	0.48
Effective gain	G	dB	-4	36.02
Antenna temp.	T_A	K	300	200
RX temp.	T_R	K	3,270	300
System temp.	T_{sys}	K	3,570	500
RX merit	G/T_{sys}	dB	-39.53	9.03
Power received	P_{RX}	dBW	-113.74	-121.05
Noise density	N_0	dBW/Hz	-193.07	-201.61

Link Budget	Symbol	Units	Uplink	Downlink
Path length	R_L	km	1,944	1,944
Path loss	L_s	dB	151.29	151.57
Atmospheric loss	L_a	dB	0.5	0.5
Polarization loss	L_p	dB	1	1
Total loss	L	dB	152.79	153.07
Data rate	R	bit/s	1.92E+4	1.5E+6
E_b/N_0	E_b/N_0	dB	36.5	18.8
Bit error rate	BER	—	1E-4	1E-5
Modulation	—	—	FSK	QPSK
E_b/N_0 threshold	E_b/N_0	dB	13.4	9.6
Implementation loss	L_{imp}	dB	15	2
Required E_b/N_0	$(E_b/N_0)_{rq}$	dB	28.4	11.6
Link margin	LM	dB	8.1	7.2

FSK: frequency-shift keying.

of the spacecraft antenna needs to be low. In other words, the antenna pattern is preferred to be omnidirectional. To close the link with such a low-gain antenna, it is straightforward to reduce the path loss by lowering the frequency. This is why we often see ultrahigh-frequency (UHF) or VHF antennas for the uplink. Accordingly, the uplink data rate is often compromised for a long distance (i.e., from Earth to the moon) to close the link.

"THERE IS MORE ROOM AT THE TOP"

It is seen from (7) that a wider channel bandwidth supports a higher data rate. Although the fractional bandwidth Δ of an antenna remains the same when scaling it from a lower frequency (e.g., UHF) to a higher band (e.g., the X band), the bandwidth $B = \Delta \cdot f_0$ is widened because of the increase of the center frequency f_0 . Hence, some use the phrase "There is more room at the top" to describe the potential of high-frequency downlinks.

CUBESAT LINK BUDGET EXAMPLES

Three link budget examples for different orbits are presented in this section. While the first two examples are based on a former CubeSat mission and practices in the SmallSat community, the third link budget includes detailed notes for readers who wish to exercise formulas developed in the preceding sections of this article.

LINK BUDGET OF AN LEO CUBESAT

This is a link budget developed by Dr. Charles Swenson for the DICE mission [25]. As seen, the antenna temperature and RX physical temperature were taken as the same in this example. The LNA temperature is taken as the noise temperature of the ground RX because the LNA gain is 30 dB, high enough to ignore the contribution from the second stage. The 3.08-dB noise figure of the LNA and the Earth temperature were used to calculate the LNA noise temperature.

The CubeSat RX temperature was calculated using the RX sensitivity (17), where the sensitivity was -96 dBm, and the required E_b/N_0 was 9.62 dB (from the required BER) plus the demodulation implementation loss of 15 dB. It should be noted that such an implementation loss is quite high, and it was an estimation to give enough margin for the link budget. The author kept the original number for historical accuracy.

In Table 3, it is seen that the data rate for the TT&C is lower than for the downlink. This makes sense because that is the link where the mission data are transmitted back to the ground, and a faster rate is preferred. With generous margins, both links "close."

LINK BUDGET OF A GEO CUBESAT

The following example is for a satellite placed farther in Earth orbit, GEO [16]. Being farther away from Earth compared to LEO, the physical temperature of the spacecraft was taken as slightly less than the Earth temperature.

The worksheets for computing noise temperatures are provided in Table 4.

SPACECRAFT RX T_{SYS}

For the spacecraft RX, the antenna temperature was estimated by summing 25% of the Earth and 75% of the cold sky temperature. This is because the antenna is a dipole with a 90° half-power beamwidth; therefore, one quarter of its pattern (i.e., the beam front) looks at Earth and the rest (the side and back) at the sky. At the uplink frequency 145.8 MHz, the coldest sky temperature is 355 K, due to galactic noise (10).

Studying the radio sky map [24], it is seen that a GEO spacecraft appears along the horizontal line ($y = 0^\circ$) and

passes through the warmest area of the sky ($t_c \geq 28$ K) for about 2 h in the vicinity of the 19th hour. Therefore, the sky temperature was calculated from (10) with $t_c = 28$ to ensure that the link is closed for 22 h except for when the spacecraft is in the warmest zone. An interested reader can easily verify that the link is marginal for the hottest sky. One could either lower the data rate (such as 900 bit/s) to close the uplink or choose a different modulation method (such as GMSK) that yields a lower E_b/N_0 threshold.

The RX temperature was computed from (15) and (16) with the noise figures being 1.03 dB and 4.45 dB for the LNA and second stage, respectively. The T_{sys} was then computed with a simplified treatment of the antenna efficiency as 100%.

TABLE 4. GEO LINK BUDGET.

Design Element	Symbol	Units	Uplink	Downlink
Frequency	f	MHz	145.8	437.45
TX power	P_{TX}	Watts	10	2
TX power	P_{TX}	dBW	10	3
TX power	P_{TX}	dBm	40	33
TX	Symbol	Units	Uplink	Downlink
Antenna gain	G_t	dB	18.5	2.2
Line A length		m	1	0.075
Line B length		m	0.3	0.075
Line C length		m	25	0.3
Line loss/m		dB	0.05	0.58
Total line loss	L_l	dB	1.315	0.261
Connector loss		dB	0.05×6	0.05×4
Filter insertion loss	IL	dB	1	1
Other inline loss	L_{other}	dB	0.5	0.5
Antenna mismatch loss	RL	dB	0.5	0.24
Loss total		dB	3.62	2.2
Power transmitted	P_t	dBW	24.89	3.01
RX	Symbol	Units	Uplink	Downlink
Antenna gain	G_r	dB	2.2	18.5
Line a length		m	0.2	2.5
Line b length		m	0.1	0.3
Line c length		m	0.1	0.3
Line loss/m		dB	0.4	0.092
Total line loss	L_l	dB	0.16	0.29
Connector loss		dB	0.05×6	0.05×4
RX	Symbol	Units	Uplink	Downlink
Filter insertion loss	IL	dB	1	1.5
Other inline loss	L_{other}	dB	0.5	0
Loss total		dB	1.96	1.99
Line efficiency	e_L		0.6368	0.5
Antenna temp.	T_A	K	895	500
Spacecraft temp.	T_p	K	280	290
LNA temp.	T_{LNA}	K	75	72
LNA gain	G_{LNA}	dB	18	18
Second-stage temp.	$T_{2\text{nd}}$	K	500	1,000
System temp.	T_{sys}	K	764.6	510
Link Budget	Symbol	Units	Uplink	Downlink
Ground pointing loss	$L_{\theta G}$	dB	0.5	0.5
Polarization loss	L_p	dB	0.1	0.1
Path loss	L_s	dB	167	176.5
Atmospheric loss	L_a	dB	2.1	2.1
Ionospheric loss		dB	0.7	0.4
Spacecraft pointing loss	$L_{\theta s}$	dB	4.7	0.3
Data rate	R	bit/s	1,800	300
E_b/N_0	E_b/N_0	dB	17.7	16.2
Bit error rate	BER		$1\text{E}-4$	$1\text{E}-5$
Modulation			FSK	GMSK
E_b/N_0 threshold	E_b/N_0	dB	13.4	9.6
Implementation loss	L_{imp}	dB	1	0
Required E_b/N_0	$(E_b/N_0)_r q$	dB	14.4	9.6
Link margin	LM	dB	3	6.6

GROUND STATION RX T_{sys}

The antenna temperature was measured by the process explained in the “Noise and System Noise Temperature” section. The measured noise level was -132.4 dBm, and the bandwidth of the instrument was 10 kHz. Then, the warmest galactic noise temperature 84 K was added to have a more conservative estimation.

The noise figures for the LNA and second stage were 1 dB and 6.6 dB, respectively. The same simplification for T_{sys} was employed as was done for the spacecraft RX.

The final link budget is copied as follows. The line efficiency e_L for (6) was taken as the combined line attenuation and was computed from $10^{-(\text{total line loss})}$. It is seen in this example that the downlink data rate is lower than the TT&C's. This is because of the larger path loss for the downlink frequency. Increasing the gain of the ground antenna is one way to solve the problem,

although fitting high-gain antennas to CubeSats is an ongoing research interest [5].

LUNAR ORBIT CUBESAT $\leftrightarrow$ EARTH

This example is an analysis performed by the author for a link between a CubeSat on the lunar orbit and the Earth ground station. The example presents budgets for up- and downlinks with separate sets of tables to make room for detailed descriptions or formulas so that a reader could easily verify the example data. An average moon–Earth distance of $R_L = 382,500$ Km is used to calculate the path loss. The abbreviation *Temp.* is used for temperature. The uplink and downlink budgets are shown in Tables 5 and 6.

It should be noted that the BER for the uplink, which is for commanding the spacecraft, is taken as more critical (i.e., a lower tolerance for errors) because the command link needs to be

TABLE 5. LUNAR UPLINK BUDGET.

TX	Symbol	Value	Descriptions
Frequency	f	450 MHz	
Wavelength	λ	66.7 cm	
TX power	P_{TX}	12.6 W	
Power (dBW)	P_{TX}	11.0037 dBW	$10 \log_{10}(P_{\text{TX}})$
Line loss	L_l	2.5 dB	Σ Loss (lines A, B, and C)
Filter loss	IL	1 dB	
Other loss	L_{other}	0.5 dB	Directional coupler
Σ Inline loss	L_T	4 dB	$L_l + IL + L_{\text{other}}$
Gain	G_t	39.5 dB	
Space loss	L_s	197.1587 dB	$20 \log_{10}(\lambda/4\pi R_L)$
Pointing loss	L_θ	2 dB	See note 1
Polarization loss	L_p	0.5 dB	
Atmospheric loss	L_a	1 dB	
Σ Link loss	L_{link}	200.6587 dB	$L_s + L_\theta + L_p + L_a$
RX	Symbol	Value	Descriptions
Line loss	L_l	0.02 dB	Σ loss (lines a, b, and c)
Filter loss	IL	1 dB	
Other loss	L_{other}	0.5 dB	Hybrid
Σ inline loss	L_t	1.52 dB	$L_l + IL + L_{\text{other}}$
Line efficiency	e_L	0.7047	$10^{-0.1 \times L_t}$
Gain	G_r	1.8 dB	
RX power	P_{RX}	-153.8749 dB	$P_t + \Sigma G_{r,t} - \Sigma L_{\text{link},t}$

Noise	Symbol	Value	Descriptions
Antenna temp.	T_A	290 K	See note 2
Physical temp.	T_p	290 K	See note 3
LNA gain	G_{LNA}	18 dB	
LNA temp.	T_{LNA}	160 K	See note 4
System temp.	T_{sys}	500 K	See note 5
Noise power/Hz	N_0	-201.6115 dB	$10 \log_{10}(k_B T_{\text{sys}})$

Link Budget	Description	Value
R	Data rate	1,200 bit/s
BER	Bit error rate	$1\text{E}-7$
Modulation	QPSK	Not coded
E_b/N_0		16.9448 dB
$(E_b/N_0)_{\text{th}}$	Threshold	11.3 dB
L_{imp}	Implementation loss	1 dB
Link margin	$E_b/N_0 - (E_b/N_0)_{\text{th}} - L_{\text{imp}}$	4.6 dB

- Note 1:* Pointing loss includes both the ground and spacecraft antennas' pointing loss. It should be noted that pointing loss can be different for up- and downlink and is related to locations of the CubeSat with respect to Earth. Only the location where the CubeSat is directly over the ground station is considered in this example.
- Note 2:* Since the uplink antenna on the spacecraft is an omnidirectional dipole, the antenna temperature can be simplified as an average of the Earth and sky temperature with galactic noise included. This average value is lower than the Earth temperature. In this budget, the Earth temperature is entered to leave enough margin.
- Note 3:* The physical temperature of the spacecraft RX, including the front end. This is taken to be the same as Earth's instead of a cooler lunar temperature to be more conservative to accommodate any unforeseen noise.
- Note 4:* This is estimated from (16) by taking a noise figure of 1.55, a fairly modest value for an LNA.
- Note 5:* This is found by rounding the value computed from (13) and (15) by taking the antenna's efficiency as 95% and the second-stage RX temperature as 1,000 K.

TABLE 6. LUNAR DOWNLINK BUDGET.

TX	Symbol	Value	Descriptions
Frequency	f	8 GHz	
TX power	P_{TX}	2 W	
Power (dBW)	P_{TX}	3.0103 dBW	$10 \log_{10}(P_{TX})$
Line loss	L_l	0.03 dB	Σ loss (lines A, B, and C)
Filter loss	IL	1 dB	
Other loss	L_{other}	0.5 dB	Directional coupler
Σ inline loss	L_T	1.53 dB	$L_l + IL + L_{other}$
Gain	G_t	22 dB	
Link loss	Symbol	Value	Descriptions
Space loss	L_s	222.16 dB	$20 \log_{10}(\lambda/4\pi R_L)$
Pointing loss	L_θ	2 dB	See note 1
Polarization loss	L_p	0.5 dB	
Atmospheric loss	L_a	1 dB	
Σ link loss	L_{link}	226.66 dB	$L_s + L_\theta + L_p + L_a$
RX	Symbol	Value	Descriptions
Line loss	L_l	2.52 dB	Σ loss (lines a, b, and c)
Filter loss	IL	1 dB	
Other loss	L_{other}	0.5 dB	Hybrid
Σ inline loss	L_t	4.02 dB	$L_l + IL + L_{other}$
Line efficiency	e_L	0.3963	$10^{-0.1 \times L_t}$
Gain	G_r	64.5 dB	
RX power	P_{RX}	-142.6959 dB	$P_t + \Sigma G_{r,t} - \Sigma L_{link,t}$
Noise	Symbol	Value	Descriptions
Antenna temp.	T_A	50 K	See note 6
Physical temp.	T_p	290 K	See note 7
LNA gain	G_{LNA}	18 dB	
LNA temp.	T_{LNA}	160 K	See note 8
System temp.	T_{sys}	408 K	See note 9
Noise power/Hz	N_0	-202.5 dB	$10 \log_{10}(k_B T_{sys})$
Link Budget	Description	Value	
R	Data rate	2.5E + 5 bit/s	
BER		1E-6	
Modulation	Concatenated conv. and Reed-Solomon QPSK	See note 10	
E_b/N_0		5.82 dB	
$(E_b/N_0)_{th}$	Threshold	2.5 dB	
L_{imp}	Implementation loss	1 dB	
Link margin	$E_b/N_0 - (E_b/N_0)_{th} - L_{imp}$	2.32 dB	

- Note 6: This can be evaluated from (9) with the front hemisphere of the antenna facing cold sky and the back side being Earth. A rough estimation is taken as that the minor lobe level is -8 dB (a fairly high estimation), and all minor lobes point to Earth. The galactic noise at 8 GHz is lower than 3 K. Accordingly, 50 K is estimated to be the antenna temperature.
- Note 7: The physical temperature of the ground RX, including the front end. It is the same as the Earth temperature.
- Note 8: This is estimated from (16) by taking a noise figure of 1.55, a fairly modest value for an LNA.
- Note 9: This is found by rounding the value computed from (13) and (15) by taking the antenna's efficiency as 95% and the second-stage RX temperature as 3,000 K, a conservative estimation.
- Note 10: Concatenated conv. ($R = 1/2$, $K = 7$) and Reed-Solomon (255, 223) coding was chosen here.

accurate, whereas the errors in the downlink data can be corrected using more advanced signal processing at the ground station.

It is seen that the downlink budget is relatively marginal. This can be solved by either using a coding method that offers a lower E_b/N_0 threshold or a larger RX antenna. It is becoming a common practice that CubeSat teams rent services from ground station providers, such as NASA, and therefore two different ground antennas can be chosen for transmitting and receiving. As a reference, options for parabolic ground RX antennas are listed in Table 7 while keeping all of the rest of the parameters the same, as in this Lunar CubeSat example. The aperture efficiency of the antennas is taken as 60%.

CONCLUSIONS

This article is intended to provide a tool for those who are looking into CubeSat projects and determining spacecraft

TABLE 7. SIZE OF GROUND STATION ANTENNAS.

Diameter	Frequency	Gain	Link Margin
26 m	Uplink	39.5 dB	4.6 dB
26 m	Downlink	64.5 dB	2.32 dB
30 m	Uplink	41 dB	6.1 dB
30 m	Downlink	66 dB	3.82 dB

antenna parameters, i.e., the frequency, bandwidth, and gain, for the required or available data rate. Relations among the data rate, SNR, and system noise temperature are reviewed and explained. A clear description of how to input these values into the Friis transmission equation is presented. The main

elements of a CubeSat link budget are explained in detail, and three real-world examples are included for those who are interested in creating or reading a link budget. The system noise temperature, an entry to the link budget that is often challenging for an antenna engineer, is explained, and common simplified estimations are provided.

As a summary, a link budget sets guidelines for CubeSat antennas. The requirement for the frequency and gain pattern of the CubeSat antenna can be different for up- and downlink. While the uplink antenna is for command and tracking and is preferred to have an omnidirectional pattern, the downlink is desired to offer a high data rate, which results in a need for high-gain antenna. For the ground station antenna, properties such as the minor lobe level affect the antenna noise temperature. The impedance matching, polarization, and transmission lines in the TX contributes to the path loss. The pointing loss is determined from the gain pattern, location, and orientation of the spacecraft. All of these are entries to the link budget and the resulting parameters that close a link become design tasks for antenna engineers.

The examples in this article are for frequencies below the X band. It should be noted that Earth's atmospheric noise becomes substantial for millimeter waves [22, Figure 12-4], and effects due to water and atmospheric gas have to be carefully considered.

Although the examples are for CubeSats, the link budget development outlined can be modified for general communication links. A major modification would be the antenna temperature, and the Earth temperature may be taken for both the TX and RX antennas. Earth temperature may vary depending on surrounding radiation sources or landscapes, and one could look up published statistical numbers for the entry to be more accurate. Literature on the system noise temperature is relatively scattered and inconsistent, and the equations presented in this article have been carefully verified.

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